



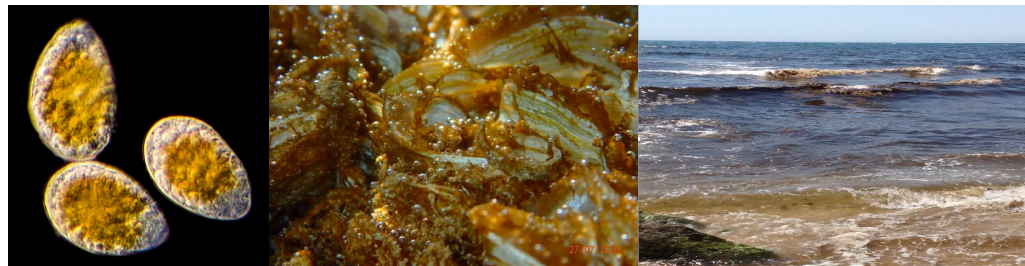
Microscale Ocean Biophysics 8.0

27 May-5 Jun 2026 Cargèse, Corsica. (France)

Procedure for sampling of benthic microalgae

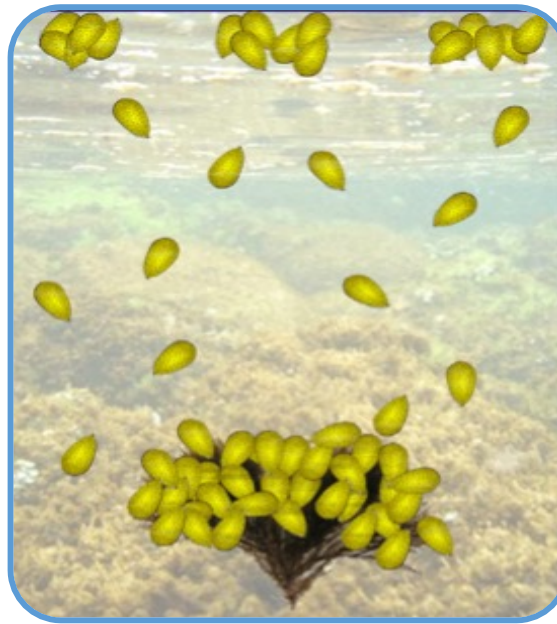
With focus on *Ostreopsis* and harmful microalgae

Elisa Berdalet (ICM-CSIC)



Some microalgae have a tychoplanktonic behaviour

- floating aggregates
- in the water column
- **attached in the benthos: main reservoir**



(Anne-Sophie Pavaux, adapted from Mangialajo et al. 2017)

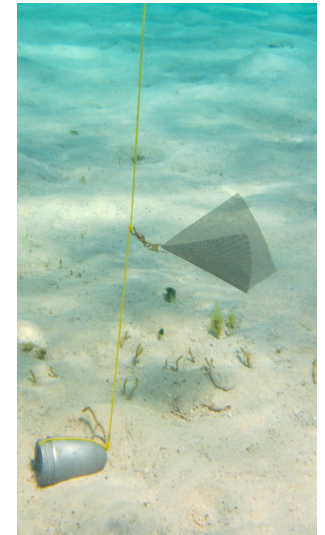
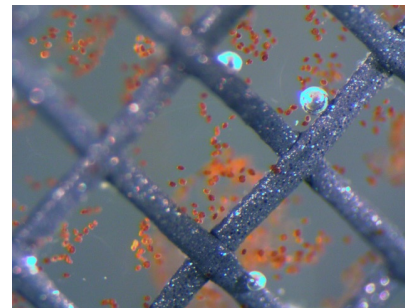
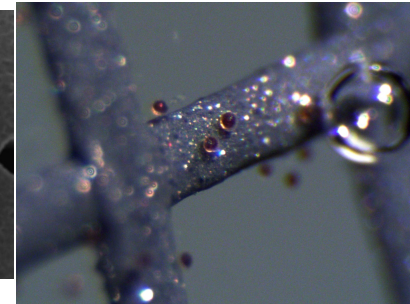
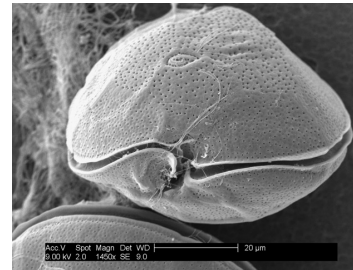
Thus, we may need to sample both plankton and benthic substrates

Traditionally, macroalgal sampling



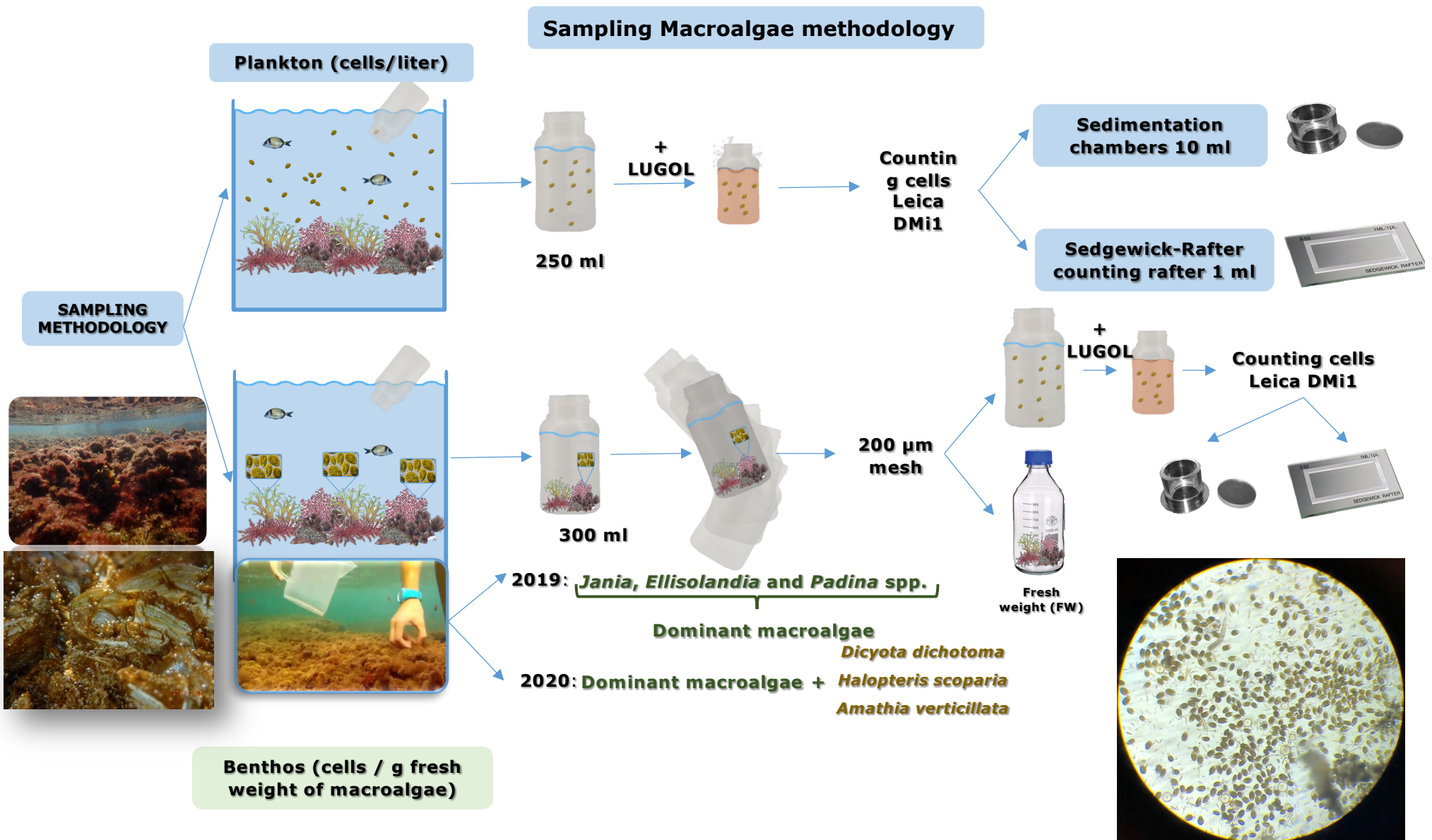
Artificial substrate, Tester et al 2014:

- suggested standardized method
- needs more research, site-specific
- useful to explore presence of BHAB species, molecular tools



Traditional sampling

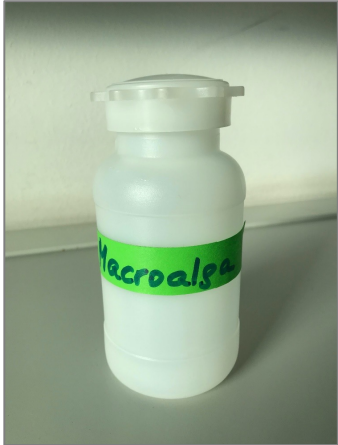
Slide 4



Source: N.I. Medina-Pérez (PhD thesis)

Materials for scientific and monitoring purposes, and for citizen science

1. Plastic bottle (250 mL) to collect macroalgae in seawater
2. A 200 μm filter unit
3. Plastic graduated jar (with volume marks)
4. Plastic forceps
5. Lugol's iodine (neutral or acidic) and plastic Pasteur pipettes (pipettes are used to dispense the Lugol's iodine)
6. A 50 mL plastic sample tube to hold the Lugol's iodine preserved sample; it should adjust in the grid holder or the chosen beaker
7. Parafilm to seal the sample tubes
8. Paper towel, markers, pencil, tape, 20 cm (approx.) plastic ruler
9. Resealable zipper storage bags to keep the macroalgae
10. Underwater snorkelling gear (glasses, tube and booties)
11. Electrical conductivity meter
12. Notebook with excel sheet table (Use a pencil to fill in)



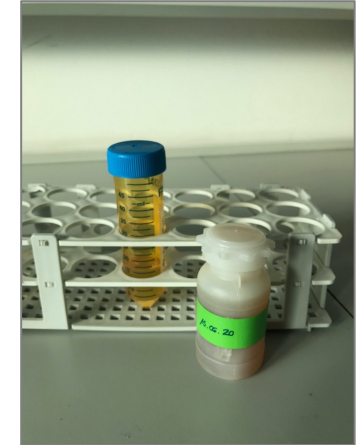
1. Plastic bottle (250 mL) to collect macroalgae in seawater



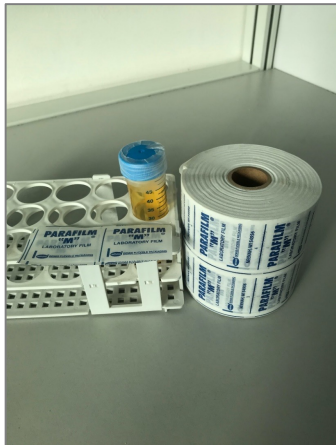
2. A 200 µm filter unit
3. Plastic graduated jar (with volume marks)
4. Plastic forceps



5. Lugol's iodine and Pasteur pipettes



6. A 50 mL plastic tube to hold the Lugol's iodine preserved sample; it should adjust in the grid holder. Alternative plastic beaker.



7. Parafilm to seal the sample tubes



8. Paper towel, markers, pencil, tape, 20 cm (approx.) plastic ruler
9. Resealable zipper storage bags to keep the macroalgae



10. Underwater snorkelling gear (glasses, tube and booties)



11. Electrical conductivity meter

12. Example of Notebook

Location	Sampling code	Date & Hour [dd/mm/yy & 24 hour clock]	Water temperature (°C)	Salinity	Meteo Conditions	Sampled Macroalgae or substrate	Percolated water volume (mL)	Photo #
Cargèse 1	MOB.8_1	29/06/2026 09:00 hr	21,0	37.8	Sunny	<i>Jania</i> 60% <i>Padina</i> 40%	220	1, 2, 3
Cargèse 1	MOB.8_2	29/06/2026	21,2					
Cargèse 1	MOB.8_3	29/06/2026	21,4					
Cargèse 2	MOB.8_4	29/06/2026	21,6	37.9	Sunny	Pebbles	240	4, 5
Cargèse 2	MOB.8_5	29/06/2026	21,6					
Cargèse 2	MOB.8_6	29/06/2026	21,6					

- a) Take a photograph of the sampling site (Photo 1)
- b) Explore the area covered by macroalgae. If this is your first time sampling, find out what the dominant macroalgae are in the beach and collect three macroalgal samples at three spots located within 10 m of each other. Always record the macroalgae you collect (see NoteBook, on Slide 7).

Collect about 5 - 10 g (handful, see Photo 2) of **macroalgae** and put them into the 250 mL plastic bottle (Photo 3). The macroalgae should be gently grabbed from the holdfast (looks like a root), avoid touching the frond (looks like leaves and stem) part that contains the attached microalgae we are looking for. Fill at least two thirds of the bottle with seawater.

If there are no macroalgae present at the sampling site, please collect rocks, pebbles or sand, and record the approximate surface area sampled in cm^2 .



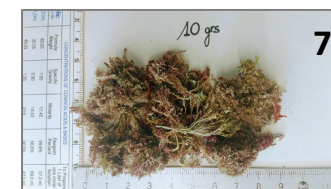
- c) Close the bottle and **shake it intensively for approximately 60 seconds**. This will dislodge the microalgae we are looking for that are strongly attached to the macroalgae without damaging them.
- d) Pour the content of the flask through the 200 μ m mesh and collect it in a jar (Photo 4). The macroalgae will be retained on the sieve (Photo 5). The seawater now contains *Ostreopsis* and other microorganisms (Photo 5). Using a plastic forceps, hold the macroalgae in the sieve and using the forceps press them to help complete water drainage. If some macroalgae remain in the bottom of the 250 mL plastic bottle, use more seawater to rinse them into the sieve again.
- Record the volume of seawater:** We need this number to calculate the microalgae, *Ostreopsis*, cell concentration. Record it the Notebook (slide 7)
- e) The brown colour of the water can be due to a mixture of *Ostreopsis*, diatoms, other dinoflagellates and flagellates. When *Ostreopsis* is at high cell concentrations, brown filaments are observed at the water surface (as in Photo 5). If the sample contains sand, the sand will sink rapidly to the bottom of the jar.



Ostreopsis Macroalgae

- f) Using the forceps, mix the water in the jar and try to disperse the *Ostreopsis* aggregates (if any are present). Now fill the 50 mL plastic tube (as in Photo 6 or alternative sampling bottle provided) with the *Ostreopsis*-containing percolated water sample collected. Add some drops of Lugol's iodine into the 50 mL plastic tube, close the lid and invert a few times, check that it has a whisky-like colour. Add more Lugol's iodine to get the desired colour. Label the tube/beaker with the following details using the marker provided: location, sampling date (day/month/year) and the hour the sample was collected. Use parafilm or plastic cling film to wrap and seal the 50 mL plastic tube lid. This helps to prevent leakage and sample loss when it is transported to the laboratory for analyses.
- g) With a paper towel, dry the macroalgae collected and place it on a white piece of paper with ruler (see Photo 7). Write down the location, date and the hour and take a photograph.

Please Note: the concentration of the benthic microalgae are expressed by weight of the macroalgae. If the sample is delivered to the laboratory on the same day of sampling, weigh the macroalgae (fresh weight) and record the weight in grams (write details in the Notebook, slide 7). Alternatively, put the macroalgae into a plastic bag (with label including location, date and hour) and keep the macroalgal sample frozen until it is delivered to the laboratory to get weighed.



- h) It is important to observe the environment. Look out for foam, floating aggregates, water discolorations, "fleurs d'eau" (see Photo 8) and take photographs. While out sampling, if you observe discoloured water or something unusual, it is recommended that you collect a water sample and/or the floating aggregate directly into a plastic bottle. Please add a few drops of Lugol's iodine and take plenty of photographs of the unusual event.
- i) Because *Ostreopsis* and other benthic microalgae cover the benthos (seabed) with a brown mucilaginous film, take pictures underwater if possible (Photo 9).
- j) If possible, measure temperature and salinity and record the meteorological conditions.



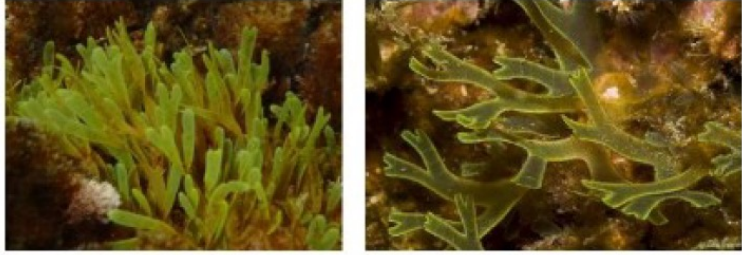
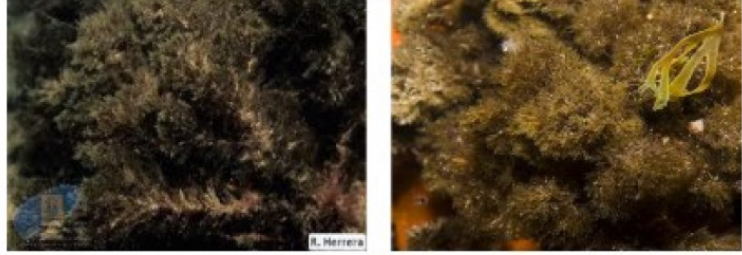
SUMMARY

- a) Take a photograph of the sampling site (Photo 1)
- b) Take 1 or 3 macroalgal samples, representative of the dominant macroalgal species in the beach.
- c) Carefully collect about 10 - 20 g (handful size) of macroalgae and put them into the 250 mL plastic collection bottle.
Fill about two thirds of the collection bottle with seawater. If there are no macroalgae present at the sampling site, please collect rocks, pebbles or sand, and record the approximate surface area sampled in cm².
- d) Close the collection bottle and shake it intensively for approximately 60 seconds to dislodge the microalgae (*Ostreopsis*) from the macroalgae.
- e) Pour the content of the flask through the 200 µm mesh and collect it in the jar (Photo 4). The macroalgae will be retained on the filter unit (Photo 5) and the seawater now contains *Ostreopsis* and other microorganisms (Photo 5). Using a plastic forceps, hold the macroalgae in the filter unit and use the forceps to press the macroalgae and ensure all the water is drained through the filter unit. Note the "Water volume" (see slide7).
- f) Using the forceps, mix the water in the jar and try to disperse the *Ostreopsis* aggregates (if any are present). Fill the 50 mL plastic tube or the chosen beaker with the collected water sample containing *Ostreopsis*. Add some drops of Lugol's iodine into the 50 mL plastic tube.
- g) With a paper towel, dry the macroalgae collected and place it on a white piece of paper with ruler (see Photo 7). Write down the location, date and the hour, the macroalgae sampled, and take a photograph. Put the macroalgae into a plastic bag and keep it frozen until it is delivered to the laboratory to get weighed.
- h) Record the information in the notebook.

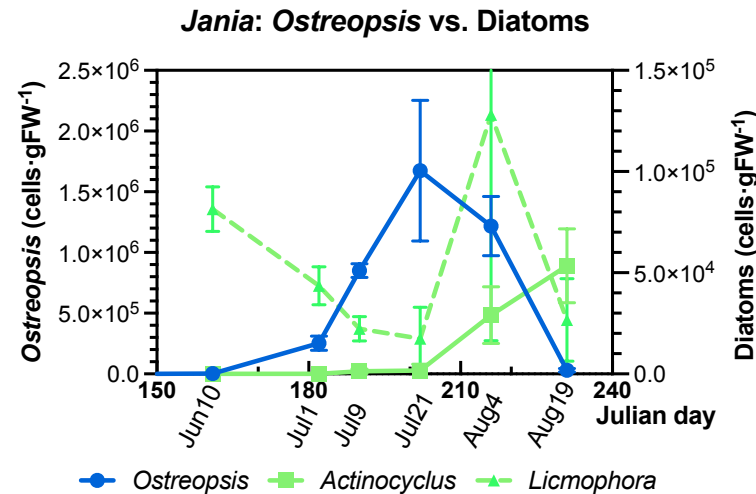
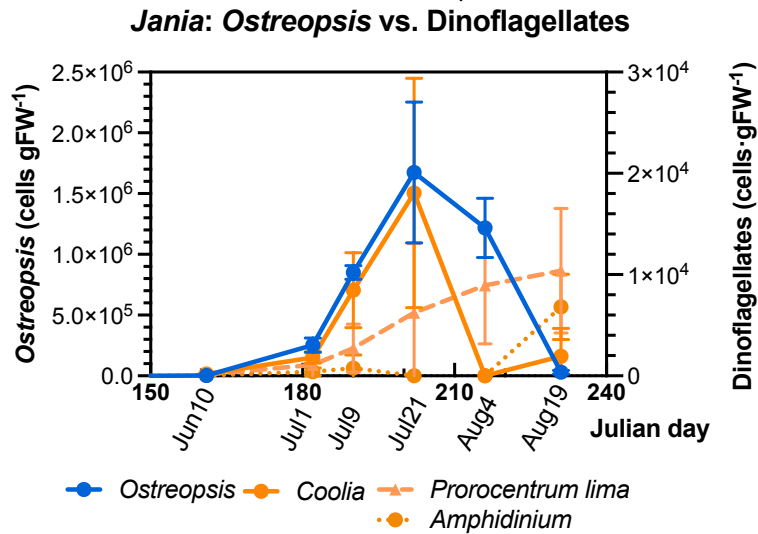
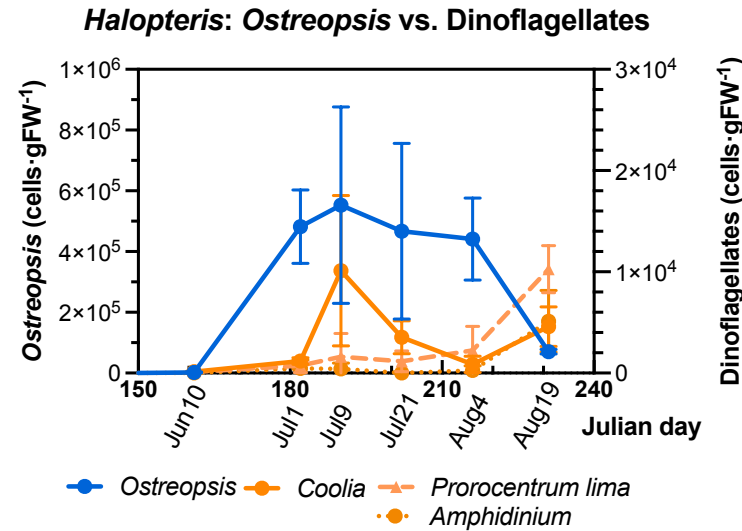
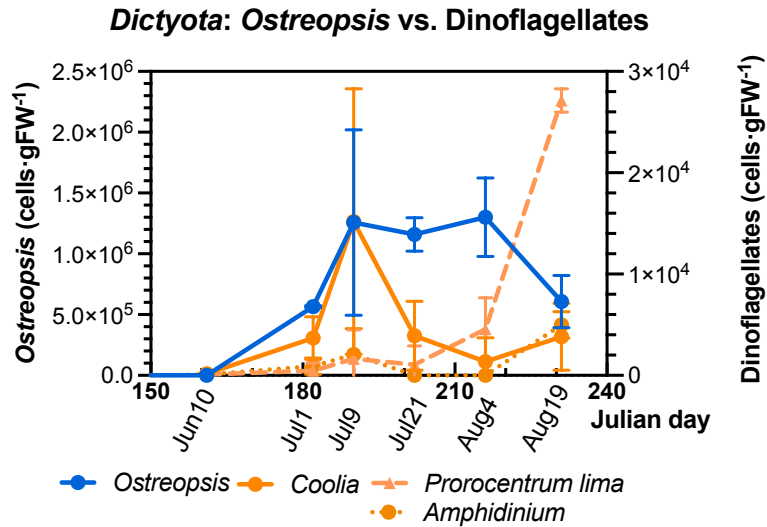
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C. Cusack (MI, Ireland), R. Lemée (LOV, France)

Red algae

Brown algae

ALGUES VERMELLES	ALGUES BRUNES
<u><i>Jania rubens</i></u> 	<u><i>Padina pavonica</i></u> 
<u><i>Ellisolandia elongata</i></u> 	<u><i>Dictyota dichotoma</i></u> 
<u><i>Laurencia sp.</i></u> 	<u><i>Halopteris scoparia</i></u> 

Example of variability of microalgae overtime and depending on the sampled macroalgae (ECOS project, PID2022-138753OB-C21 & C22)

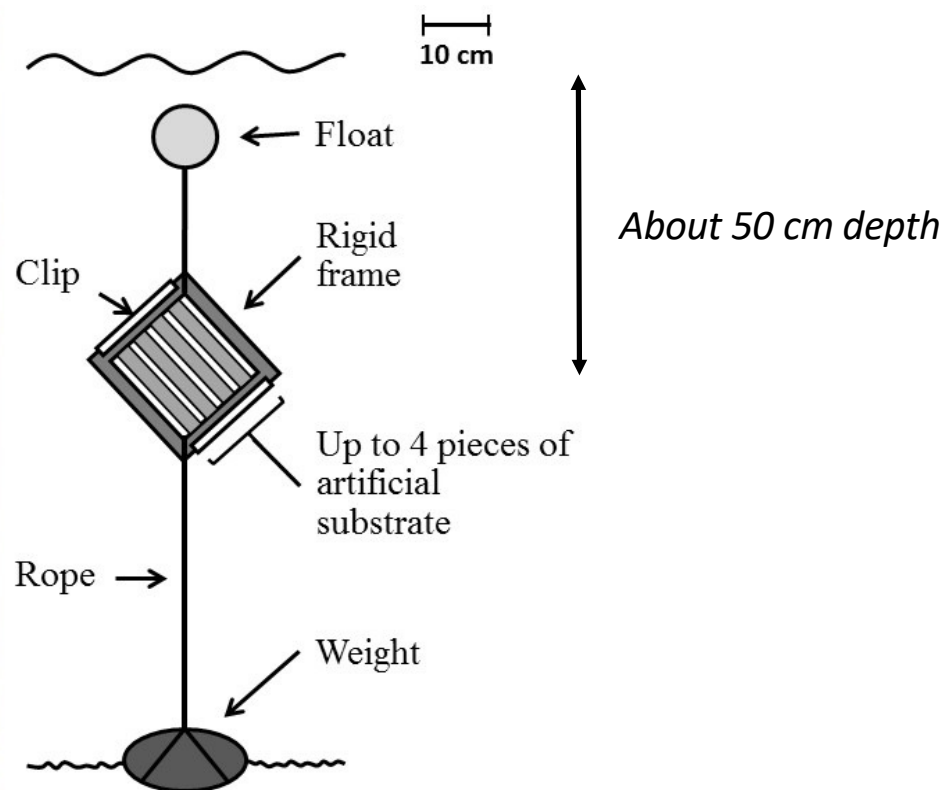
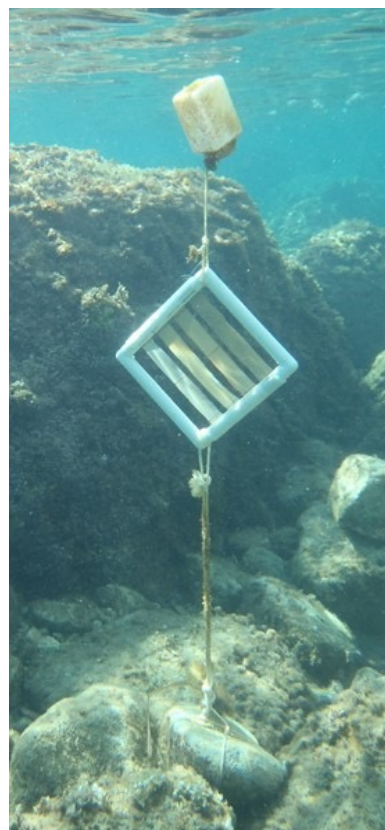


Look at the biological components:

- macroalgal succession
- microalgal succession

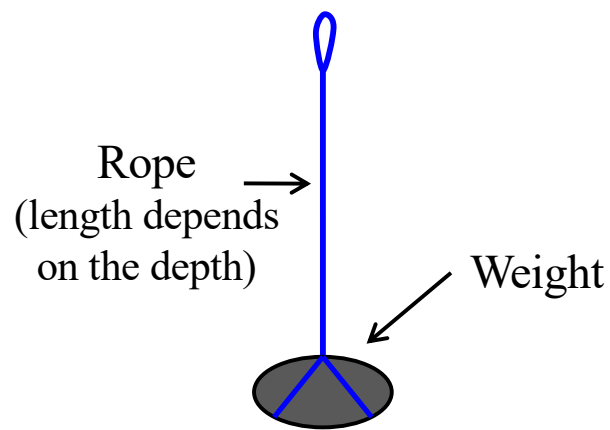
Source: S. Christian (Master thesis)

Description of the set-up for artificial substrate deployment
Project M3-HABs – Mariachiara Chiantore, Rodolphe Lemée

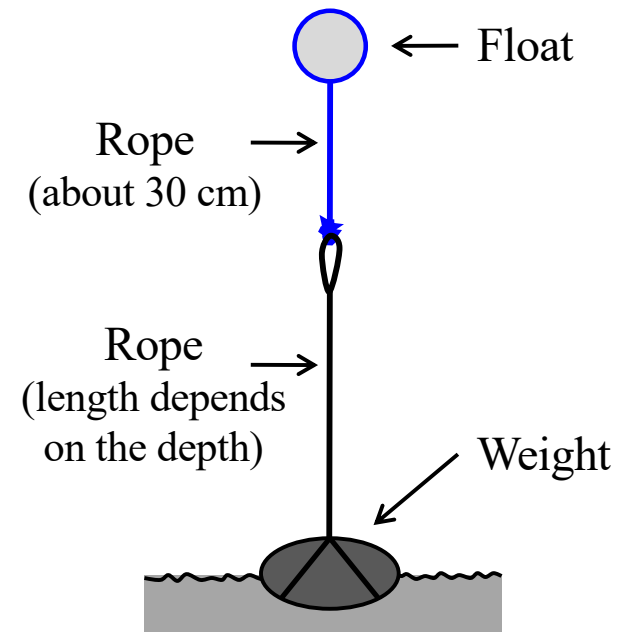


Pieces of advice for the 1st set-up

1- Fix a rope on a rock, with a loop on it.
(potentially out of the water)

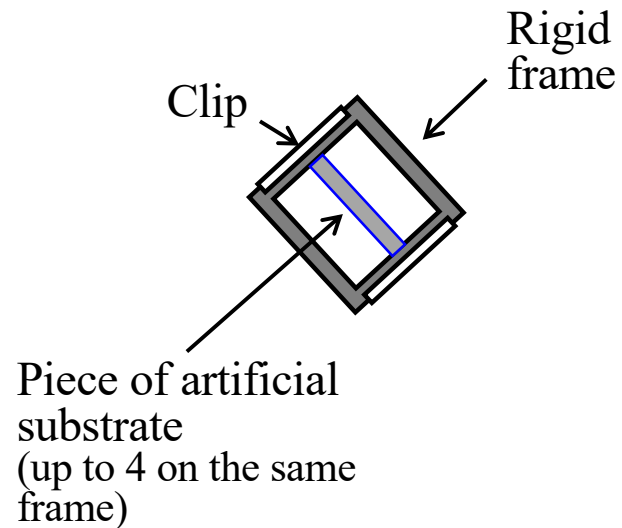


2- Fix the float on the loop using a rope, so that the loop will be easily accessible on site.
Then, **put everything in the water**, at about 30 cm from the rock where you are used to do your macroalgae sampling.

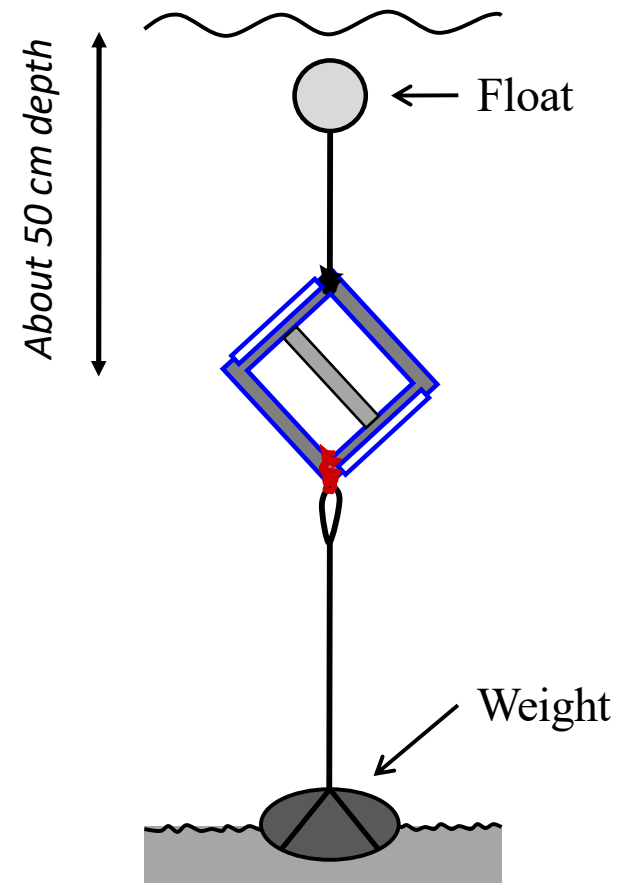


Pieces of advice for the artificial substrate deployment

3- In the lab, **fix the piece(s) of artificial substrate** on the rigid frame.
(width = 2.5 cm; Length = 25 cm)



4- One day before the “classical sampling”, **start the incubation** of artificial substrates by **fixing the frame at about 50 cm depth**. Use an additional small piece of rope for fixation of the frame on the loop.



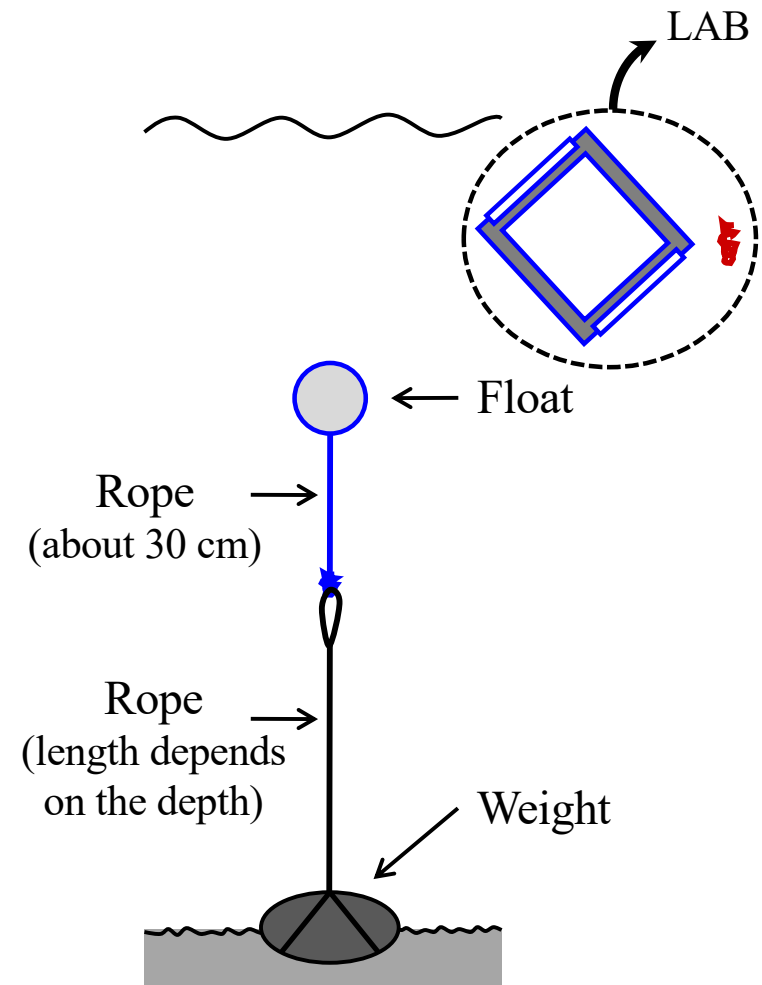
Pieces of advice for sampling of the artificial substrates

5- After 24h, **sample the artificial substrates** using scissors:

- a- cut on one side.
- b- fill the sampling bottle with surrounding seawater.
- c- gently put the artificial substrate in the bottle without touching it.
- d- close the bottle immediately underwater.

(please see the movie for a visualization of the sampling step)

6- Remove the rigid frame at put the float back, fixed on the loop. The frame can be taken back to the lab, with collected samples.



For the treatment, follow the same protocol than a macroalgae sampling

7- Back to the lab:

- a- **fix the sample** with acidic lugol
- b- **dislodge *Ostreopsis* cells** by shaking and rinsing (2 times)
- c- **keep a sub-sample** for counting (better in the fridge)
- d- **count using a 1mL- Sedgewick Rafter counting chamber**

8- For standardization of the data:

First, you have to **correct your data of abundances by the volume of washing** in order to get the number of isolated cells in the whole sampling bottle.

Then, you can **standardize your data** according to either:

a- **the cutting area** ($2.5 \times 16.5 = 41.25 \text{ cm}^2$)

-> Divide the number by **41.25 cm²**

or:

b- **the “real surface area”** = 182.57 cm^2

(estimated from the diameter of the fibers (320 μm), the porosity (1.27 mm) and the equation described in Tester et al 2014)

-> Divide the number of cells by **182.57 cm²**.